FAST STEERING MIRROR FOR LINE-OF-SIGHT STABILIZATION

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ABSTRACT

Compact Fast Steering Mirror (FSMs) mechanisms have a wide range of applications including satellite communication systems, Earth observation missions and scientific space mission. The FSM mechanism developed at CSEM is a compact design and aims to control the position and stability of a planar mirror. Compared to other mechanisms on the market, CSEM's FSM is distinguished by its relatively large mirror size, large angular stroke, and high accuracy. In addition, its versatile design allows it to be easily adapted to a wide range of applications such as satellite communication or Earth observation.

The mechanism is designed with the objective to ease the manufacturing process, accommodation, actuation, and to reduce the overall costs while achieving exceptional stability over the ±3 mrad tip/tilt mechanical range. A breadboard mechanism was produced and assembled at CSEM, replacing the 60 mm diameter SiC or Zerodur mirror by a representative Aluminium dummy. A performance measurement campaign was conducted with excellent results.

1 INTRODUCTION

The Fast-Steering Mirror (FSM) mechanism developed at CSEM (see Fig. 1) is a compact design whose purpose is to control the position and stability of a plane mirror. Compact FSMs have a wide range of applications including satellite communication systems, Earth observation missions and scientific space missions.

2 DESIGN DESCRIPTION

CSEM has designed and developed, a Fast-Steering Mirror (FSM-60) mechanism represented in Fig. 1 and Fig. 2 for applications such as satellite communications or Earth observation missions [1]. The mechanism is designed with the objective to ease the manufacturing process, accommodation, actuation, and to reduce the overall costs while achieving exceptional stability over the ±3 mrad tip/tilt mechanical range. The simplest mirror shape is circular, but an elliptical shape can be considered. The actuation of the mirror is done with the use of three voice coil actuators arranged at 120° providing a large angular stroke and versatility.

A low noise current amplifier was custom developed to ensure the highest possible accuracy of the voice coil drive current with a measured noise level below 0.2 µA. The highest sensitivity and resolution capacitive sensors were integrated with bandwidths up to 10-20 kHz to measure the mirror position which is then controlled in closed loop. A central mirror-flexure membrane supporting the mirror is implemented ensuring a friction-free tip, tilt and piston movement. The lightweighted mirror design with low mobile mass and overall compact configuration eliminates the need for an additional launch locking device.

Figure 1: Breadboard of Fast Steering Mirror, FSM-60

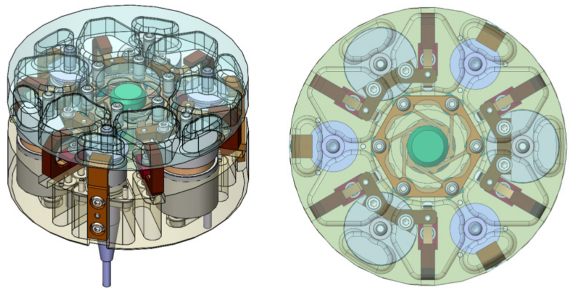

Figure 2: FSM concept

2.1 Simplified product tree of the FSM mechanism

The mechanism is composed of 4 critical sub-assemblies as shown in the exploded view of the FSM in Fig. 3.

1) **A stable planar mirror.** This is the simplest solution to address general requirements while simplifying the manufacturing process, reducing cost and simplifying accommodation and actuation interfaces. Symmetry is always simpler and cheaper to implement due to the repetition of N similar parts, where N is to be minimized. The simplest mirror shape is circular, but an elliptical shape can also be considered, as it maximizes the optical diameter taking into account the available mechanical volume.

2) **Three Voice coil actuators at 120°.** The voice coil solution allows for mechanical angular strokes up to ±4 mrad with a 58 mm mirror diameter. It can also provide for an increased stroke range if needed. Voice coil actuators can be COTS components or custom-made for optimized performances. Space-qualified voice coils can also be integrated. The voice coil solution has also the advantage of avoiding hysteresis compared to piezo actuators while drawbacks are heat generation and low intrinsic stiffness. This low stiffness needs closer evaluation but does not necessarily imply the need for a launch locking mechanism. This low intrinsic stiffness also means less chance to over constrain the mirror. The solution with three actuators has many advantages compared to four :
   - It is the minimal number of actuators allowing for tip/tilt corrections, so it does not over constrain the system (4 actuators create hyperstaticity, leading to potential deformation of the mirror and pre-stress on the flexure pivot/membrane)
   - It saves space for sensors that can then be directly mounted on the mirror back
   - It reduces mass and cost
   - Three Voice Coils (VC) along with three sensors is an elegant solution control-wise

3) **Position sensors.** The mirror position is directly measured, depending on the application needs, by three capacitive or Eddy current sensors with the mobile parts thereof attached to the mirror. Three sensors directly mounted on the mirror back allow for a direct mirror position measurement and a precise amplitude control as requested, particularly for science applications (compared to telecom applications). The sensors should ideally be mounted on the device under motion, and as close as possible to the actuator, or directly opposite to the actuator, hence direct coupling of the sensor actuator pair is straightforward. Eddy current or capacitive sensors exhibit the best resolution for direct measurement of the mirror position. Strain gauges require strain items, hence rigid actuators such as PZT or the variant APA, while magnetic actuators tend to use non-contact magnetic or electric type sensors.

4) A **central mirror-flexure** supporting the mirror and ensuring the tip/tilt and piston movements starting from a neutral (0, 0, 0) position and applying a restoring force for any displacement along these three degrees of freedom. This solution allows for maximizing first non-desired resonant frequency and ensures sufficient stiffness. Flexures have low hysteresis, are reproducible, have no resolution limit, no stick slip, no wear (other than fatigue failure), no lubricants, compared to ball or contact/sliding joints. Nevertheless, flexures have a limited angular stroke, and a restoring force opposing the displacement from a neutral position, both positive and negative depending on the application.

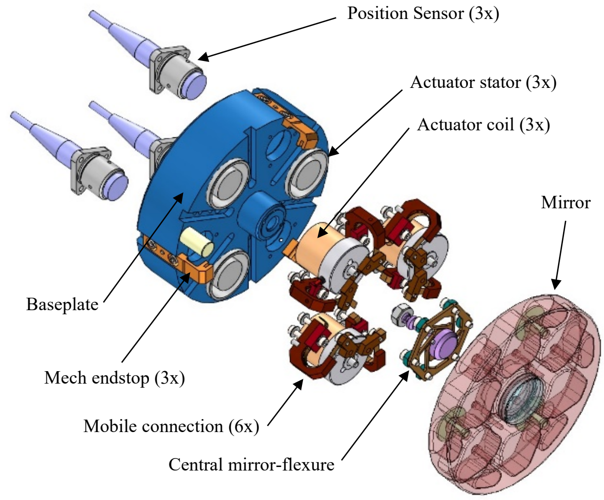

Figure 3: FSM-60 concept exploded view

2.2 Mirror design

The mechanism is designed with the goal of controlling the position and stability of a planar mirror (Fig. 4). This design facilitates the manufacturing process, accommodation and actuation while reducing overall costs. For the sake of simplicity in terms of manufacturing and control for breadboard activities, a flat circular mirror of dimensions 60 mm in diameter was fabricated for the FSM-60 breadboard.

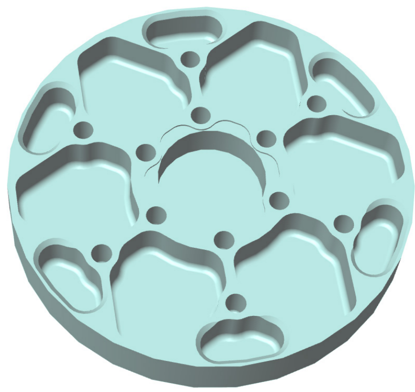

Figure 4: Circular mirror 60 mm in diameter

The positions and dimensions of the mirror ribs have been optimised to increase the stiffness, accommodate the holes for fixing the actuators and sensor plates through glued threaded inserts, while keeping the overall mass as low as possible. The rib layouts and dimensions can vary in accordance with the chosen material (e.g. Zerodur or Silicon Carbide).

2.3 Optical Mirror Interfaces

To interface the mirror with the FSM structure, a monolithic, CuBe flexure membrane is implemented to support the mirror ensuring the tip/tilt and piston movements starting from a neutral position (Fig. 5). The membrane applies a restoring force for any displacement along three degrees of freedom. This solution allows for maximizing first tip/tilt resonant frequency and ensures enough stiffness. Flexures have low hysteresis, are reproducible, have no resolution limit, no stick slip, no wear and no lubricants. The interface between the central axis and the structure is screwed in a stud bolted to the I/F structure.

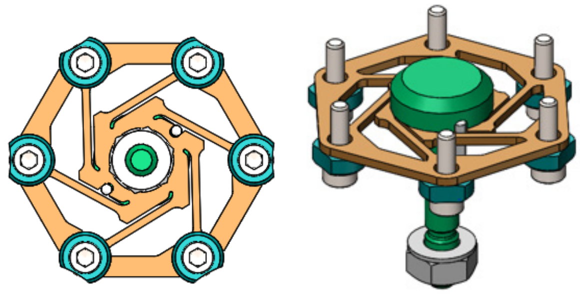

Figure 5: Flexure membrane with assembly bolts and central tie rod

2.4 Position sensors technology

The capacitive and the Eddy current technologies are best suited for the FSM, but as the specification [1] demanded for resolutions better than 0.3 µrad, clearly the capacitive sensors were the best choice, despite their higher sensitivity to air pressure, vacuum and humidity, and somewhat lower signal bandwidth. These effects can be managed by proper calibration with the measurement chain.

Highest sensitivity and resolution capacitive sensors with bandwidths up to 10-20 kHz measure the mirror position which is then controlled in closed loop. The selected sensor is the model Micro-Epsilon *CSH02-Cam1,4* with nominal calibration corresponding to 0.2 mm full sensor range (FSO) and 4 nm resolution at maximum @ 8.5 kHz bandwidth.

The extreme resolutions request for the FSM of the nm levels for typically > 100 µm travel range requires, furthermore, electronics with the highest excitation voltage to increase the signal-to-noise ratio of the desired sensor signal. A commercial version of such an electronic system from Micro-Epsilon (DT 6530 sensor system) was used for the development.

2.5 Actuators

The actuation of the mirror is done with the use of three voice coils arranged at 120° (Fig. 6). Voice coils allow for large angular strokes of up to 4 mrad with a 58 mm diameter mirror.

As a baseline COTS solution, Moticont voice coils LVCM-016-013-01 were integrated in the FSM-60 breadboard. These actuators are a minor variant (different stroke) of a model that has been qualified by CSEM on the CLUPI instrument for the ExoMars rover for a focus mechanism (linear scanner) that incorporated COTS Moticont voice coils [2] & [3].

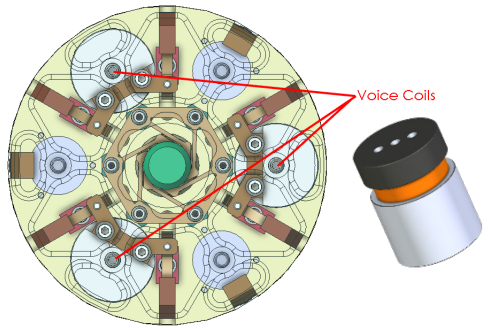

Figure 6: Three voice coil actuators positioned at 120°

In order to balance the inertia of the system, the actuators act at about 2/3 of the mirror radius. For a mirror of 60 mm in diameter, this corresponds to a 20 mm action radius. Therefore, a total stroke of 112 µm is necessary to cover a tip/tilt mechanical correction range of ±3 mrad.

2.6 Sensor and actuator accommodation

The sensor targets are directly glued on the back of the mirror using inserts made of Invar to match Zerodur thermal expansion coefficient. The baseline for the sensor target material is Aluminium. The fixation of the sensor head in the base plate uses a sleeve as shown in Fig. 7 (left). The gap between the sensor head and the target is set at 0.12 mm but can be adapted to the application.

The actuator coil is fixed to the mirror in a similar method as the sensor targets. The heavier stator part of the actuator with the magnets is attached to the mechanism housing thus reducing the mirror mobile mass.

To limit the mechanism motion and avoid unwanted collisions, three end-stops are implemented. They are made of plastic and act on the sensor targets by limiting their vertical motion (red rectangle). An illustration is given in Fig. 7.

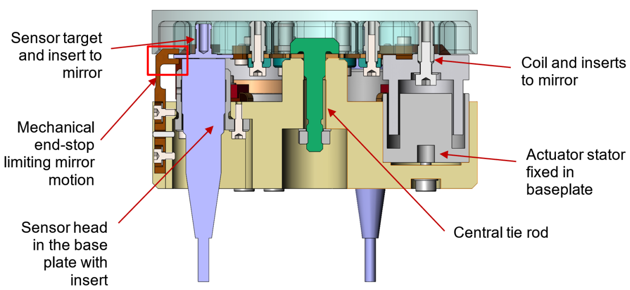

Figure 7: Cross-section view of the mechanism

2.7 Voice coil mobile connections

Electrical power is transmitted to the actuators with the use of mobile connections. Their design is optimized to reduce the mass and the exported forces and torques that may be generated. The dimensions of the rolling ribbons are large enough to provide the necessary current while limiting their heating. The radius of curvature is sized to provide infinite lifetime cycles.

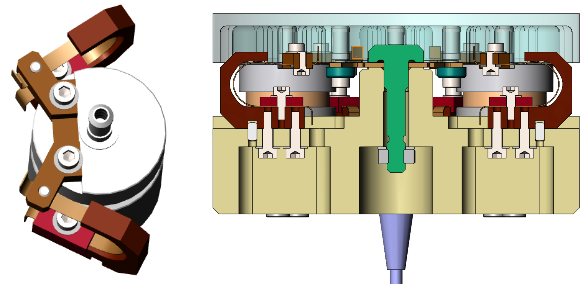

Figure 8: Voice coil with mobile connections

A static, non-linear analysis was performed for a typical load case with tip and tilt angles of 1.5mrad (mechanical). The maximum stress in the mobile ribbon does not exceed 90 MPa for a maximal vertical displacement of about 40 µm of the connection while the maximum stress on the membrane is below 230 MPa. No severe stress concentrations are observed for the 1.5 mrad mechanical tip and tilt rotations neither on the flexure structure nor on the ribbon structure. These stress values are below the infinite fatigue life of the material.

3 ANALYSIS

3.1 Modal analysis

A finite element analysis of the mechanism was performed and the computed eigenmodes were very close to those that were measured with the FSM prototype during the performance tests. The table below shows the first frequencies calculated and the corresponding measurements were Rx/Ry (Tip/tilt): 137 Hz, Z: 440 Hz (Piston), see §5.2.

Table 1. FEM Modal analysis results

Frequency [Hz]	
136.5	RY
136.6	RX
464.6	z
999.5	Rz

3.2 Random vibration

A preliminary analysis of the FSM with typical launch loads was performed by imposing a random vibration profile through a base excitation node linked to the baseplate. A random vibration spectrum corresponding to an input load of 24 Grms in X/Y and 43 Grms in Z was used to evaluate the response of the mechanism. The FEM model for the random vibration analysis in Fig. 9 is shown with the parts contributing to the moving mass (mirror, actuator coils, mobile connections, flexure membrane and sensor targets).

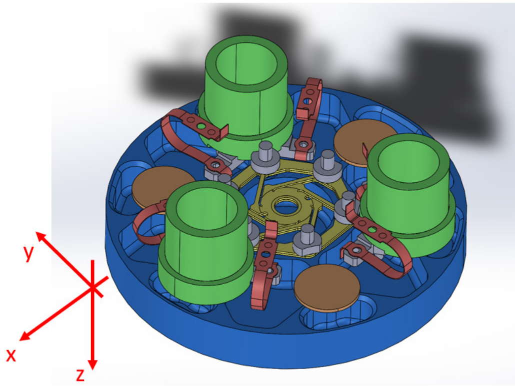

Figure 9: Random vibration model

The results for the Z random spectrum are presented, since they give the highest displacements and stresses, corresponding to the worst-case scenario, with the Z random vibration spectrum aligned normal to the FSM planar mirror.

Z random spectrum

The random vibration spectrum applied in the Z direction of the FEM model is compared to the PSD Z displacement of the mirror at the end-stop position in Fig. 10.

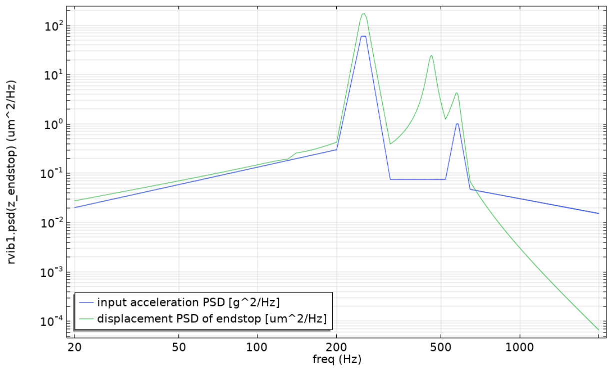

Figure 10: Input random acceleration (blue) compared to PSD Z displacement of the mirror at the end-stop position (green)

The 137 Hz eigenmodes are not excited, but the 464 Hz mode is. The RMS displacement is $1\sigma = 119$ µm (Fig. 11) and the RMS Von Mises stresses are about $1\sigma = 600$ MPa (Fig. 12).

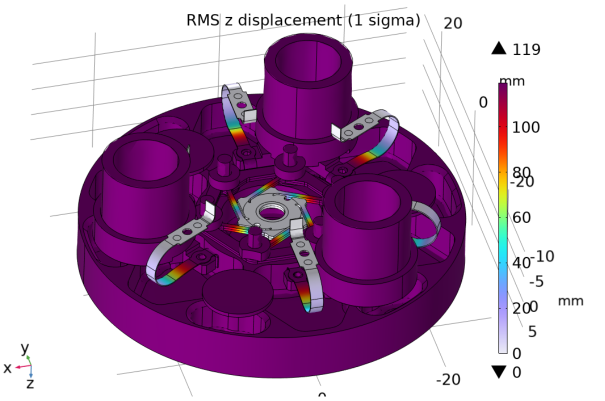

Figure 11: RMS Z-displacement

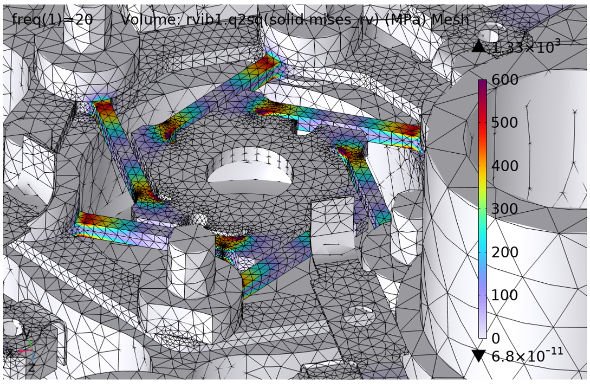

Figure 12: RMS Von Mises stresses

Table 2: Random vibration displacement and stresses

Applied direction on model	RMS Z displacement of endstop		Von Mises RMS stress	
	1σ	3σ	1σ	3σ
	[µm]	[µm]	[MPa]	[MPa]
X/Y	31	93	55	165
Z	119	357	600	1800

The cells highlighted in red give the largest displacement at the mirror end-stop and von Mises RMS stress in the CuBe membrane flexure arms (3σ values).

In the worst-case scenarios, displacements in excess of ±120 µm mean that, in the absence of stops, the mirror sensor targets will collide with the sensor heads, otherwise the stress in the membrane arms could exceed the elastic limit of CuBe (approx. 1200 MPa). The integrated end-stops will prevent this and preserve the integrity of the mechanism. Further analysis would be needed to refine this scenario in greater detail at the level of the end-stops. However, no major difficulties are foreseen regarding the compatibility of the mechanism with typical launch loads.

3.3 Thermal analysis

A preliminary thermal analysis was performed based on preliminary specifications for a foreseen Earth Observation application. An example of the temperature environment for the non-operation case corresponding to worst-case limits of +60°C and -30°C was considered for analysis. Given these temperature ranges, no criticalities at component level (actuators, sensor, mirror, membrane and mechanical parts) are foreseen.

Effect of temperature variations, due to CTE difference between materials will induce stress in the membrane tangential arms and mobile connections. This is accounted for in part by design and sizing. The load case for +60°C is presented in Fig. 13. Considering that if the mechanism is assembled at 20°C, the mean stress that we observe at 60°C with a SiC mirror, is <100 MPa which is totally acceptable. This is a conservative, preliminary analysis, considering that there is no surface sliding between different CTE materials in contact with each other.

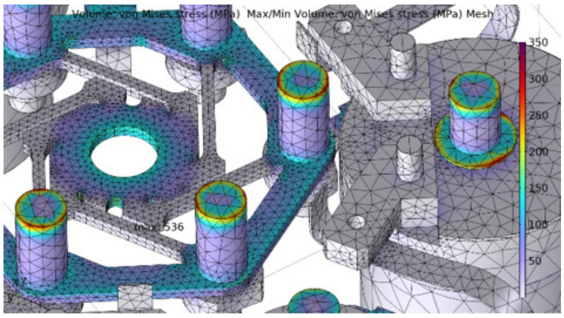

Figure 13: Von Mises stresses in the CuBe membrane and mobile connections at 60°C

4 ELECTRONIC GROUND SUPPORT EQUIPMENT (EGSE)

The performance tests with the FSM were performed using Electronic Ground Support Equipment (EGSE) shown in Fig. 14 and comprised of:

- FSM mechanism (blue box)
- Digital controller (black box)
- Voice-coil amplifier (red box)
- Sensor electronics (green box)

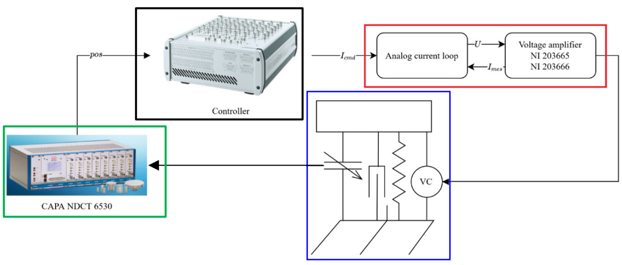

Figure 14: Electronic ground support equipment for tests

4.1 Voice coil electronics

The electronics to power the voice coil actuators is sized so that it is possible to control the force moving the mirror with the requested resolution. Angular resolution requirements are the main sizing requirement. It translates to the need for a low noise current amplifier capable of controlling current at µA level. A low noise current amplifier was custom developed to ensure the highest possible accuracy of the voice coil drive current with a measured noise level below 0.2 µA. The amplifier developed by CSEM (Fig. 15) was used to power the FSM breadboard during the performance tests.

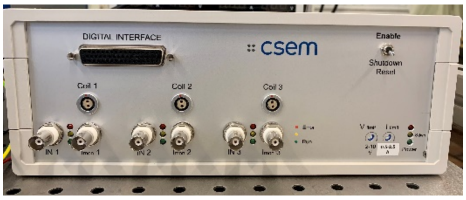

Figure 15: Low-noise 3-channel voice-coil amplifier electronics

4.2 Capacitive sensor electronics

The FSM breadboard prototype tests were performed using the highest resolution industrial capacitive system proposed by Micro-Epsilon, namely the capaNCDT DT 6530 rack system with up to 8 sensor demodulators.

Figure 16: Micro-Epsilon capacitive sensor electronics

This system is based on the highest-voltage excitation electronics for the optimum signal-to-noise ratio. However, the -3 dB bandwidth is set to 8.5 kHz and the digital readout is limited to 7.8 ksamples/sec. This system has both an analogue output filtered at 8.5 kHz (-3 dB), and a digital readout having gone through a further calibration and linearization process. The selected sensor is the model *CSH02-Cam1,4* with nominal calibration corresponding to 0.2 mm full sensor range (FSO). For a future space application, specific sensor electronics would need to be developed using the Micro-Epsilon electronics as a reference for the specifications.

4.3 Controller

Performance tests with the controller were performed at CSEM using the dSpace MicroLabBox controller hardware running a Simulink model. The controller is comprised of three main elements (Fig. 17):

- State controller
- Feed forward
- Observer

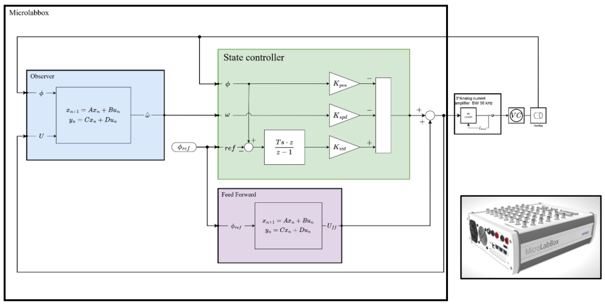

Figure 17: Controller schematic

5 TESTING

5.1 FSM breadboard with dummy mirror

A breadboard mechanism was produced and assembled at CSEM, replacing the flat circular SiC or Zerodur mirror by a representative Aluminium dummy with the same mass and inertia for the mobile assembly. A smaller, optical mirror mounted within the dummy mirror was used to perform the tests with an autocollimator.

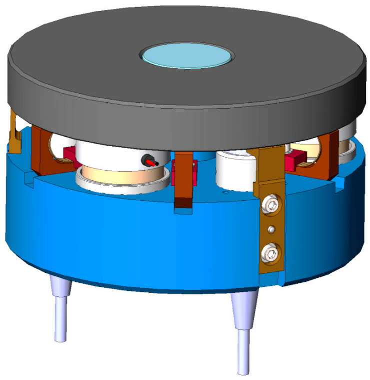

Figure 18: Breadboard FSM with dummy mirror

5.2 Frequency

Injecting a sine sweep on coil 1, while keeping the current at 0 on coil 2 and 3 leads to the response depicted in Fig. 19 for s1 to s3. Similar curves are obtained when using coil 2 or 3. The observed modes are the main tilting mode at 137 Hz and Z mode at 441 Hz.

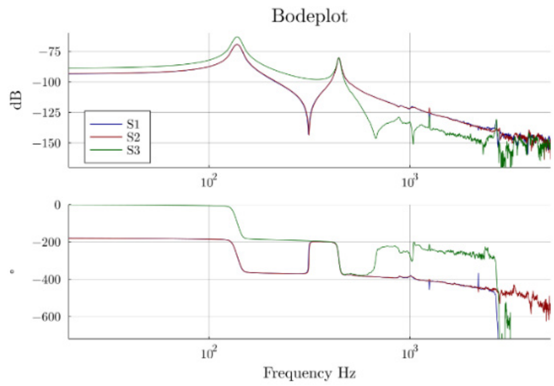

Figure 19: One channel sine sweep excitation

5.3 Test results

A trajectory scanning the working area was performed where the actual measured angles are recorded with an autocollimator. The displacement error after recentering and rotational alignment is depicted in Fig. 20, the maximum error is 33 µrad or a non-linearity of 1.7%. After correction with a calibration table the linearity error drops to 2.3 µrad or 0.12 %

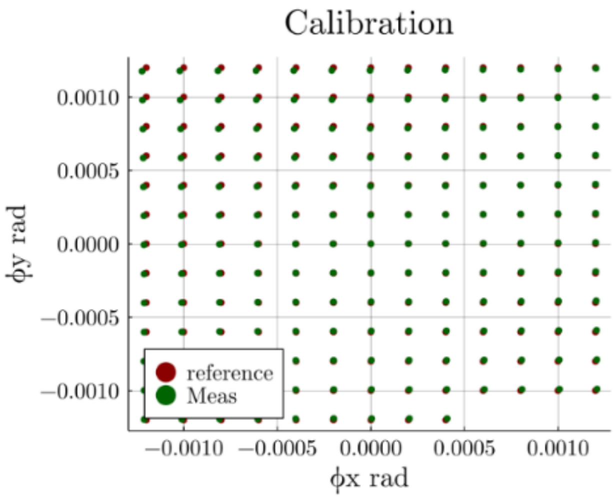

Figure 20: Calibration chart

Another important transfer function depicted in Fig. 21 is the closed loop response to a reference command. It basically shows that the control is effective up to 200 Hz, thus, a signal with a 200 Hz BW (Bandwidth) can be followed.

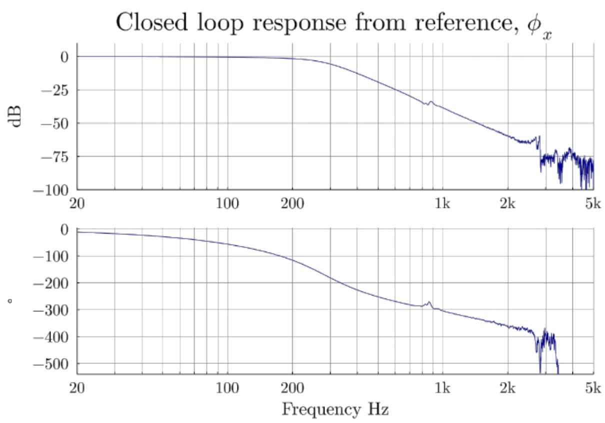

Figure 21: Closed loop response

6 FSM PERFORMANCE CHARACTERISTICS

The properties and performance measurements on the FSM-60 breadboard mechanism is summarized in Table 3.

Table 3: FSM-60 performance summary

Characteristics	FSM performance
Mechanical angular range (Funct/Max)	±3 mrad / ±4 mrad
Number of actuators / type	3 / Voice coil
Sensor type	Capacitive
Mirror diameter	60 mm
Typ. Resolution in closed loop	0.34 µrad
Typ. Stability (RMS)	0.02 µrad up to 100Hz 0.16 µrad at 7'800 Hz
Typ. Repeatability	1 µrad
Dimensions	Ø60 mm x 45 mm
Mass	<200g with Zerodur mirror
Stiffness in Z	0.508 N/µm
Resonant frequency	130 Hz
Operation Voltage/control signal	±10 V
Operating current	±2.5 A

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